**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 203 · Karnaugh maps and simplification

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Simplify ¬(A ∨ B).

Q2. Keys 7 and 12 hash to index 2 in a table. What is that called?

Q3. What is 1 XOR 1?

Q4. If A changes from 0 to 1 while B is always 1, which variable is constant?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Karnaugh map — Grid for simplifying Boolean functions using adjacent combinations.

Gray code — Ordering in which adjacent binary labels differ by one bit.

group — A rectangle containing 1, 2, 4, 8, … adjacent 1-cells.

constant variable — A variable with the same value in all cells of a group.

wraparound — The opposite edges of a Karnaugh map are adjacent.

## New learning

Read the labels — Rows represent A, and columns use Gray order BC: 00, 01, 11, 10.

Group neighbouring ones — The four 1s form one 2×2 rectangle. Adjacency across rows is allowed.

Keep what is constant — A and C change through the group, but B remains 0. So the simplified result is ¬B.

## Worked example

Four 1s fill the entire A=0 row. Simplify the output.

1. Predict and explain — B and C each vary across the row; neither is needed in the expression.

2. Check and extend — A stays 0 in the whole group. The simplified term is ¬A.

# We Do · Practise with guidance

Four 1s fill columns BC=11 and BC=10 across both rows. Simplify.

```
BC columns: 00 01 11 10
A=0:        0  0  1  1
A=1:        0  0  1  1
```

Predict and explain

A changes between rows, and C changes between columns 11 and 10.

Check and extend

B remains 1, so the simplified term is B.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Why do we group neighbouring 1s in a Karnaugh map?

### Question 2

The whole A=1 row contains four 1s. Which simplified term remains?

### Question 3

Columns BC=00 and BC=10 are joined by wraparound adjacency. Four 1s occupy both these columns. Simplify.

## Exit Ticket · Answer before checking

A group of four 1s can remove two changing variables.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. ¬A ∧ ¬B.

Q2. A collision.

Q3. 0.

Q4. B, at value 1.

### You Do pages · Independent questions

Q1. To simplify the Boolean expression by removing variables that change inside a group.

Q2. A. Both B and C change within the row, while A remains 1.

Q3. ¬C. A changes between the rows and B changes between 00 and 10; C stays 0.

### Exit Ticket · Model answer

True. The group leaves only variables constant across all four cells.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**